Which phase of compiler does NOT use symbol table?

a) Code generation
b) Syntax Analysis
c) Lexical Analysis
d) None of the other options

Ans: d

2.

Which phase of compiler is Syntax Analysis?

a) First

b) Second

c) Third

d) None of the mentioned

Ans: b

3.

Output of the syntax analysis is called

a) Parse tree
b) Keyword tree
c) Binary tree
d) All of the other options

Ans: a

4.

A programming language does not allow integer division operation. This is generally detected in the phase of

a) Lexical Analysis

b) Syntax Analysis

c) Semantic Analysis

d) None of the other options

Ans: c

5.

Which of these is not true about Symbol Table

a) All the labels of the instructions are symbols

b) Table has entry for symbol name address value

c) Perform the processing of the assembler directives

d) Created during pass 1

Ans: c

6.

A compiler can check

a) Logical error

b) Syntax error

c) Both logical error and syntax error

d) Not logical and syntax error

Ans: b

7.

Error recovery helps to

a) Report multiple errors
b) Rectify multiple errors
c) Both report and rectify multiple errors
d) None of the other options

Ans: a

8.

Converting a hardware description into actual circuitry is known as

a) Silicon Compilation
b) HDL Compilation
c) Circuit Compilation
d) None of the other options

Ans: a

9.

Loops are the major targets for optimization since

a) Loop may go to infinite execution
b) Loop body is repeated to several times
c) Condition check takes exceedingly large time
d) None of the other options

Ans: b

10.

For maximum speed of execution of target code , temporary variables be best allocated to

a) Swap space
b) Main memory
c) CPU registers
d) None of the other options

Ans: c

11.

Intermediate code helps in

a) Program Analysis
b) Code optimization
c) Retargeting code
d) Code check

Ans: c

12.

Output file of Lex is ____ the input file is Myfile

a) Myfile.e
b) Myfile.yy.c
c) Myfile.lex
d) Myfile.obj

Ans: b

Week 2 ASSIGNMENT

1.

A regular expression represents

a) Part of a language

b) Cannot represent any language

c) Constituent strings of a language

d) None of the other options

Ans : c

2.

When expression sum=3+2 is tokenized then what is the token category of 3

a) Identifier
b) Assignment operator
c) Integer literal
d) Addition operator

Ans : c

3.

For the Fortran language statement "DO 5 I = 1.25" returns token IDENTIFIER for DO 5 I after looking upto

a) I

b) =

c) .

d) 5

Ans : c

4.

Which of the following are Lexemes?

a) Identifiers

b) Constants

c) Keywords

d) All of the mentioned

Ams : d

A regular expression for accepting strings with exactly one 1 more than 0's is

a) 0*1

b) (0|1)*1(0|1)*

c) (0|1)*1(0|1)*|1(0|1)*

d) Not Possible

Ans : d

Which one of the following languages over the alphabet {0,1} is described by the regular expression: (0+1)*0(0+1)*0(0+1)*

a) The set of all strings containing the substring 00.

b) The set of all strings containing at most two 0's.

c) The set of all strings containing at least two 0's.

d) The set of all strings that begin and end with either 0 or 1.

Ans : c

Finite automata is an implementation of

a) Regular expression

b) Any grammar

c) Part of the regular expression

d) None of the other options

Ans : a

The automation which allows transformation to a new state without consuming any input symbols:

a) NFA
b) DFA
c) Pushdown automata
d) All of the mentioned

Ans : a

The _____ of a set of states, P, of an NFA is defined as the set of states reachable from any state in P following ε-transitions

a) ε-closure

b) ε-park

c) Q in the tuple

d) None of the mentioned

Ans : a

Between NFA and DFA which one is powerful

a) NFA
b) DFA
c) Equally powerful
d) Cannot be said definitely

Ans : c

Subset Construction method refers to

a) Conversion of NFA to DFA

b) DFA minimization

c) Eliminating null references

d) ε-NFA to NFA

Ans : a

Which of the following do we use to form an NFA from a regular expression

a) Subset construction method

b) Powerful set construction method

c) Thompson construction method

d) Scott construction method

Ans : c

A certain compiler corrects errors like "fi" to "if" automatically. This is an example of recovery in

a) Panic mode
b) Delete character
c) Replace Character
d) Transpose character

Ans : d

Week 3 ASSIGNMENT

1.

Which of the following is a lexical analysis tool

a) Lex
b) Flex
c) Jflex
d) All of the other options

Ans : d

2.

In a lex specification file "?" stands for

a) 0 or more occurrences of preceding regular expression
b) One or more occurrences of preceding regular expression
c) Exactly one occurrences of preceding regular expression
d) None of the other options

Ans : d

3.

Between NFA and DFA which an has the potential to have more states in it
a) DFA
b) NFA
c) Cannot be said deterministically
d) Both of same number of sets

Ans : a

4.

Which of the following has more expressive power

a) DFA
b) NFA
c) Regular expression
d) All of the mentioned

Ans : d

5.

Lex specification file sections are demarcated by

a) %
b) {%
c) %}
d) %%

Ans : d

6.

ε-closure of a state includes

a) All states reachable from it by ε transitions only
b) All states reachable from it by single ε transitions only
c) All states from which this state can be reached using ε transitions
d) All states from which this state can be reached using ε transitions and all states reachable from it by ε transitions only

Ans : a

7.

Which among the following is false?
ε-closure of a subset S of Q is:

a) Every element of S
b) For any q ϵ ε-closure, every element of δ (q, ε) is in ε-closure.
c) No other element is in ε(S)
d) None of the mentioned

Ans : d

8.

Output of lex program is available in a file named
a) Lex.c
b) Lex.yy.c
c) Lex.l
d) Lex.yy.l

Ans : b

9.

All the regular languages can have one or more of the following descriptions:
i) DFA ii) NFA iii) Regular Expressions
Which of the following are correct?

a) i only
b) ii , iii
c) i , ii
d) iii only

Ans ; b

10.

Number of sections in a lex program is

a) 1
b) 2
c) 3
d) 4

Ans : c

Week 4 ASSIGNMENT

1.

Words of a language constitute

a) Set of terminals
b) Set of non terminals
c) Set of both terminals and non terminals
d) None of the other options

Ans : a

2.

The grammar {E →E + E | E * E | id} is

a) Ambiguous
b) Unambiguous
c) Partially ambiguous
d) None of the other options

Ans :a

3.

Which of the following are always unambiguous

a) Producing one left-most and one right-most derivation
b) Producing one left-most but may be multiple right-most
c) Producing one right-most but may be multiple left-most
d) None of the other options

Ans :a

4.

A language that admits only ambiguous grammar:

a) Inherent Ambiguous language
b) Inherent Unambiguous language
c) Context free language
d) Context Sensitive language

Ans : c

5.

A grammar with production rules { A→ Ba | Cb, B → CA, C → c | ε} contains

a) Left factor
b) Left recursion
c) Both left factor and left recursion
d) None of the other options

Ans : b

6.

For the grammar rules {S → Aa | bB, A → c |ε}, FIRST(S) is

a) {b, c}
b) {a, b}
c) {a,b, c}
d) {a ,b , c , ε }

Ans : c

7.

The grammar {E →E + T | T, T →T * F | F, F → id} is

a) Ambiguous
b) Unambiguous
c) Partially ambiguous
d) None of the other options

b

8.

Derivation produced by a top-down parser is

a) Leftmost
b) Rightmost
c) Either leftmost and rightmost
d) None of the other options

Ans : a

9.

For top-down parsing left recursion removal is

a) Mandatory
b) Desirable
c) Too complex
d) Not needed

Ans : a

10.

A grammar is ambiguous if

a) It's left most and right most derivations are different
b) More than one left most derivations exist
c) There is no left most derivation
d) There is no rightmost derivation

Ans : b

11.

A predictive parser

(A) Needs backtracking
(B) Does not need backtracking
(C) May not terminate
(D) None of the other options

Ans : a

Week 5 ASSIGNMENT

1.

In shift-reduce parsing, handle is at

a) Top of the stack
b) bottom of the stack
c) Anywhere in the stack
d) Nowhere in the stack

Ans : a

2.

Which of the following conflicts is not possible in shift-reduce parsing

a) Reduce-reduce conflict
b) Shift-reduce conflict
c) Shift-shift conflict
d) None of the other options

Ans : c

3.

Which one of the following is true at any valid state in shift-reduce parsing

a) At the bottom we find the prefixes
b) None of the mentioned
c) Stack contains only viable prefixes
d) Stack consists of viable prefixes

Ans : c

4.

For the grammar

$S \rightarrow AB \mid C$
$A \rightarrow bA \mid a$
$B \rightarrow abbS \mid bS \mid \varepsilon$
$C \rightarrow bC \mid \varepsilon$

Follow(A) is

a) a, $
b) a , b , $
c) a, b
d) b , $

Ans : b

5.

Shift reduce parsers are

a) Top down Parser
b) Bottom up Parser
c) May be top down or bottom up
d) None of the mentioned

Ans : b

6.

Shift reduce parsers are

a) Top down Parser
b) Bottom up Parser
c) May be top down or bottom up
d) None of the mentioned

Ans : b

7.

In Operator Precedence parsing handle is

a) Before <·
b) After ·>
c) Between <· and ·>
d) None of the other options

Ans : c

8.

By considering the rule B → abbS, which of the precedence relations between a and b can be inferred?

(A) a ≐ b only

(B) a ≐ b and b≐ b

(C) b≐ a and a ≐ b

(D) b≐ a and b≐ b

Ans : b

9.

An operator-precedence parser is a

a) Shift-reduce parser
b) Bottom-up parser
c) Parser constructing derivation in the reverse
d) All the other options

Ans :d

Bottom up parsing involves

a) Shift reduce
b) Handle pruning
c) None of the mentioned
d) Both shift reduce and handle pruning

Ans : d

Week 6 ASSIGNMENT

1.

For the grammar

$S' \rightarrow S$
$S \rightarrow CC$
$C \rightarrow cC \mid d$

In state 0 of LR(1) parser, an item included is

a) $C \rightarrow .cC;\ c$
b) $C \rightarrow .cC;\ d$
c) $C \rightarrow .cC;\ c,d$
d) $C \rightarrow .cC;\ c,\ \$$

Ans : c

2.

For the grammar

$S' \rightarrow S$

$S \rightarrow CC$

$C \rightarrow cC \mid d$

In state 0 of LR(1) parser, an item included is

a) $C \rightarrow .d; c$
b) $C \rightarrow .d; d$
c) $C \rightarrow .d; c,d$
d) $C \rightarrow .d; c, \$$

Ans : c

3.

In SLR parsing to get a shift-reduce conflict for state I on terminal symbol 'a',

a) $A \rightarrow \alpha.\beta$ with First(β) containing 'a' should be in I
b) $A \rightarrow \delta.$ be in I with Follow(A) having 'a'
c) $A \rightarrow \alpha.\beta$ with First(β) containing 'a' should be in I and $A \rightarrow \delta.$ be in I with Follow(A) having 'a'
d) None of the other options

Ans : c

4.

In state I we have the items $A \rightarrow \alpha.$ and $B \rightarrow \delta.$, First(A), Follow(A) and Follow(B) contains the symbol 'a'. This leads to

a) Shift-reduce conflict
b) Reduce – reduce conflict
c) Both shift-reduce and reduce –reduce conflicts
d) No conflicts

Ans : b

5.

Which of the following statements is true regarding LR parsers?

a) SLR and Canonical LR have the same number of states.
b) LALR and Canonical LR have the same number of states.
c) SLR and LALR have the same number of states.
d) All three have the same number of states.

Ans : c

6.

Construction of parsing table in which strategies do not need the Follow set?

a) SLR and canonical LR
b) Canonical LR and LALR
c) SLR and LALR
d) None of the given options

Ans : b

7.

In SLR parsing for the grammar

$E' \rightarrow E$

$E \rightarrow aEbE \mid bEaE \mid \varepsilon$

In state 0, for inputs 'a' and 'b'

a) Both will have shift-reduce conflict
b) Only 'a' will have shift-reduce conflict
c) Only 'b' will have shift-reduce conflict
d) Neither of the other options

Ans : a

8.

Amount of look ahead in LALR parser is

a) 1
b) 2
c) 3
d) None

Ans : a

9.

In SLR parsing for the grammar

$S \rightarrow B \mid SabS$

$B \rightarrow bB \mid \varepsilon$

In state 0, for inputs 'a' and 'b'

a) Both will have shift-reduce conflict
b) Only 'a' will have shift-reduce conflict
c) Only 'b' will have shift-reduce conflict
d) Neither of the other options

Ans : b

10.

Which of the following parser types is the most powerful in terms of recognizing a broader class of grammars?

a) LL(1)
b) SLR(1)
c) LALR(1)
d) LR(1)

Ans : d

11.

What is the similarity between LR, LALR and SLR

a) Use same algorithm, but different parsing table.
b) Same parsing table, but different algorithm.
c) Their Parsing tables and algorithm are similar but uses top down approach.
d) Both Parsing tables and algorithm are different.

Ans: a

Week 7 ASSIGNMENT

1.

Full form of YACC is

a) Yes Another Compiler Constructor
b) Yet Another Compiler Constructor
c) Yet Another Compiler Compiler
d) Neither of the other options

Ans : c

2.

File produced by YACC and used by Lex is SLR

a) y.tab.c
b) y.tab.h
c) y.parse.c
d) y.parse.h

Ans : b

3.

Number of sections in YACC specification file is

a) 1
b) 2
c) 3
d) 4

Ans : c

4.

In YACC rules section $$ refers to

a) First symbol on right hand side
b) Last symbol on right hand side
c) Symbol on the left hand side
d) Any of them

Ans : c

5.

Annotated parse tree means

a) Parse tree with attributes
b) Code generated
c) Correct parse tree
d) None of the given options

Ans : a

6.

For a rule "A → B b C" the symbol C will be referred to by

a) $1
b) $2
c) $3
d) $4

Ans : c

7.

Syntax directed translation helps in

a) Creating parse tree
b) Check syntactical correctness of input
c) Check if the input has foreign symbols
d) None of the othrt options

Ans : a

8.

Files generated by YACC are

a) y.tab.c, y.token.h
b) y.parse.c, y.tab.h
c) y.tab.c, y.tab.h
d) None of the other options

Ans : c

9.

Token attributes are returned by

a) yytext
b) yylen
c) yyval
d) yylval

Ans : d

10.

YYSTYPE defines the type for

a) Stack
b) Token
c) Input
d) Queue

Ans : a

11.

Which of the following statements about YACC is true?

a) YACC generates a lexical analyzer
b) YACC is used for syntax analysis
c) YACC does not support semantic actions
d) YACC does not require grammar rules

Ans : c

Week 8 ASSIGNMENT

1.

Type checking checks the input

a) Lexically
b) Semantically
c) Syntactically
d) All the other options

Ans : b

2.

If the arguments passed to a function call are such that the first and the third arguments are integers while the second one is real, the type expression for the argument list can be

a) Integer X Integer X Real
b) Real X Integer X Integer
c) Integer X Real X Integer
d) None of the other options

Ans : c

3.

Array bound check can be done

a) Statically
b) Dynamically
c) Both statically and dynamically
d) None of the other options

Ans : c

4.

Type equivalence checks whether

a) Two types expressions are same or not
b) Two expressions are same or not
c) Two statements are same or not
d) All of the other options

Ans : a

5.

Type of a statement is

a) Void
b) Type error
c) Void or type error
d) None of the other options

Ans : c

6.

Type checking done by the compiler is

a) Static
b) Dynamic
c) Both static and dynamic
d) None of the other options

Ans : a

7.

Most programming languages are weakly typed since

a) Such languages put less constraints on the programmer
b) Some type errors can be caught dynamically
c) Both of the other options
d) None of the other options

Ans : c

8.

For strongly-typed languages

a) Only static type checking is done
b) Only dynamic checking is done
c) Both static and dynamic checking are done
d) No type of checking is done

Ans : c

9.

Type casting available in many programming languages is an example of

a) Type checking
b) Type coercing
c) Type manipulation
d) None of the given options

Ans : b

10.

The type expression (Integer X Real) → (Integer → Real) corresponds to

a) A function that takes an integer and a real as arguments and returns an integer and a real
b) A function that takes an integer and a real as arguments and returns a real
c) A function that takes an integer and a real as arguments and returns an integer
d) A function that takes an integer and a real as arguments and returns a function that takes an integer and returns a real

Ans : d

11.

Type inference in programming languages refers to:

a) Explicitly specifying types for all variables
b) Automatically deducing the type of an expression at compile-time
c) Checking types dynamically during runtime
d) Ignoring type checking in a program

Ans : b

12.

Which of the following statements about type conversion is TRUE?

a) Implicit type conversion always results in data loss
b) Explicit type conversion is also called type coercion
c) Implicit type conversion is performed automatically by the compiler
d) Both (b) and (c)

Ans : c

13.

Dynamic type checking is necessary in languages that:

a) Perform all type checking at compile-time
b) Allow variables to change their type during execution
c) Do not support type inference
d) Have a very strict static type system

Ans : b

Week 9 ASSIGNMENT

1.

Self-organizing list-based symbol tables improve performance primarily due to:

a) Locality of reference in the input program
b) Locality of reference in the compiler's symbol access patterns
c) Both (a) and (b)
d) None of the above

Ans : c

2.

Most frequent operation on a symbol table is

a) Insert
b) Delete
c) Modify
d) Lookup

Ans : d

3.

Motivation behind using self-organizing list for symbol table is

a) Ease of implementation
b) Program locality
c) Insertion of symbols
d) None of the other options

Ans : b

4.

To minimize access time, symbol table should be organized as

a) Linear table
b) Tree
c) Hash table
d) Circular list

Ans : c

5.

Activation record stores

a) Parameters
b) Local variables
c) Parameters and local variables
d) Parameters , local variables and code for procedures

Ans : c

6.

Which of the following phases of compiler does NOT use symbol table?

a) Semantic analysis
b) Code generation
c) Code optimization
d) None of the given options

Ans : d

7.

If two types have same name they can be

a) Name equivalent
b) Structurally equivalent
c) Both name and structurally equivalent
d) May not be name equivalent

Ans : c

8.

Which type of compiler typically benefits from using a separate symbol table for each scope?

a) Single-pass compilers
b) Multi-pass compilers
c) Both single and multi-pass compilers
d) None of the given options

Ans : a

9.

Symbol table data is filled by

a) Lexical analyzer
b) Parser
c) Both lexical analyzer and parser
d) Neither lexical analyzer nor parser

Ans : c

10.

Which of the following is NOT likely to be kept in a symbol table?

a) Name
b) Location
c) Scope
d) None of the other options

Ans : d

11.

What is the primary purpose of a symbol table in a compiler?

a) To store machine code instructions
b) To optimize runtime performance
c) To store and retrieve identifier-related information efficiently
d) To generate intermediate code

Ans : c

12.

Which of the following data structures is best suited for managing scope information in a symbol table?

a) Stack
b) Queue
c) Linked List
d) Heap

Ans : a

13.

Which optimization technique benefits the most from information stored in a symbol table?

a) Constant propagation
b) Loop unrolling
c) Dead code elimination
d) Instruction pipelining

Ans : a

Week 10 ASSIGNMENT

1.

Which of the following is NOT possible to realize if activation record is static

(A) Passing parameters
(B) Creating local variables
(C) Supporting recursion
(D) None of the other options

Ans : c

2.

Control link points to the

(A) Current activation record
(B) Parent activation record
(C) Child activation record
(D) None of the other options

Ans : b

3.

Intermediate code generation for a compiler is

(A) Must
(B) Optional
(C) Depends on language
(D) None of the other options

Ans : b

4.

P-code is used for

(A) Stack-based machine
(B) Accumulator based machine
(C) Two operand addresses
(D) None of the other options

Ans : a

5.

In the activation record, local variables can be accessed from frame pointer with offset value

(A) Positive
(B) Negative
(C) May be positive or negative
(D) Cannot be accessed

Ans : b

6.

Access link points to the

(A) Current activation record
(B) Parent activation record
(C) Child activation record
(D) None of the other options

Ans : b

7.

If pointer is supported in the high-level language,

(A) Must also be supported in the intermediate language
(B) May not be supported in the intermediate language
(C) Depends on language
(D) None of the other options

Ans :b

8.

Frame pointer points to the

(A) Current activation record
(B) Parent activation record
(C) Child activation record
(D) None of the other options

Ans : a

9.

"Display" helps to locate easily

(A) Local variables
(B) Global variables
(C) Non-local variables
(D) All of the other options

Ans : c

10.

An intermediate language should be

(A) Close to target machine
(B) Machine independent
(C) All operators of high-level language supported
(D) All of the other options

Ans : b

11.

Which of the following is a key purpose of the stack pointer in an activation record?

(A) To store the base address of the current function
(B) To manage dynamic memory allocation
(C) To track the top of the runtime stack
(D) To store global variables

Ans : c

12.

What happens if recursion is attempted in a system with a static activation record?

(A) It executes successfully but inefficiently
(B) It results in incorrect execution since previous function calls get overwritten
(C) It executes normally without issues
(D) It leads to infinite recursion by default

Ans : b

13.

What is the primary advantage of using intermediate code in a compiler?

(A) It simplifies code generation for multiple target machines
(B) It improves the execution speed of the compiled program
(C) It eliminates the need for optimization
(D) It ensures that all high-level language features are directly mapped to assembly

Ans : a

Week 11 ASSIGNMENT

1.

For the rule S → L := E, if L is a single variable, L.place is equal to

(A) Null
(B) Some value
(C) Constant
(D) None of the other options

Ans : b

2.

For Boolean variable B, B.truelist contains

(A) List of locations at which B is true

(B) List of locations to jump to if B is true

(C) List of locations at which B is true and the locations to branch to

(D) None of the other options

Ans : a

3.

When generating code for the Boolean expression "($x \geq y$) AND ($p \neq q$)", which locations are left for back patching?

a) Falselist of $x \geq y$
b) Falselist of $x \geq y$ and falselist of $p \neq q$
c) Falselist of $x \geq y$, falselist of $p \neq q$, truelist of $p \neq q$
d) Truelist of $x \geq y$, falselist of $x \geq y$, truelist of $p \neq q$, falselist of $p \neq q$

Ans : b

4.

In three-address code, arrays are

(A) Not supported
(B) One dimensional
(C) More than one dimensional
(D) Supported via pointers

Ans : c

5.

For three address code generation of “B -> B1 or M B2”, M.quad is used to backpatch

(A) B1.truelist

(B) B1.falselist

(C) B2.truelist

(D) B2.falselist

Ans : b

6.

For the rule B -> B1 and B2, the operation "B1.false = B.false" requires two passes as

(A) B1.false is not known

(B) B.false is not known

(C) Both B1.false and B.false are unknown

(D) None of the other options

Ans : b

7.

In the rule $C \rightarrow C1$ AND NC_2 the non terminal N is used to remember the start address of:

(A) C
(B) C_1
(C) Both C_1 and C
(D) None of the other options

Ans : b

8.

In the rule S -> if B then M S N else M S, N is used to generate a jump after

(A) then-part

(B) else-part

(C) both then- and else-part

(D) None of the other options

Ans : a

9.

In the rule S -> if B then M S1, M holds the start address for

(A) S1

(B) S

(C) B

(D) None of the other options

Ans : a

10.

For three address code generation of rule “S -> while M1 B do M2 S1”, B.falselist is backpatched with

(A) M1.quad

(B) M2.quad

(C) Cannot be backpatched at this point

(D) None of the other options

Ans : c

11.

In the rule S → while M1 B do M2 S1, the non-terminal M2 is used to remember the start address of:

(A) S
(B) B
(C) S1
(D) None of the other options

Ans :c

12.

In three-address code (TAC), accessing an array element typically requires:

(A) Direct assignment without indexing
(B) Computing an address using the base address and an offset
(C) Using only registers without memory references
(D) None of the other options

Ans : b

13.

For a Boolean expression B, the attribute B.falselist contains:

(A) List of locations where B evaluates to false
(B) List of locations to jump to if B is false
(C) List of locations where B is false and the locations to branch to
(D) None of the other options

Ans : b

Week 12 ASSIGNMENT

1.

Backpatching is needed to generate intermediate code using

(A) Single pass

(B) Two passes

(C) Multiple passes

(D) None of the other options

Ans : a

2.

Jump table is suitable for

(A) Small number of cases

(B) Large number of cases

(C) Any number of cases

(D) None of the other option

Ans : b

3.

If case values are widely spaced, it is better to use

(A) Jump table

(B) Table search

(C) Either jump table or simple table

(D) None of the other options

Ans : b

4.

Function call actions are divided into sequences

(A) Calling and return

(B) Calling and composition

(C) Return and composition

(D) None of the other options

Ans : a

5.

Evaluation of actual parameters is done by

(A) Callee

(B) Caller

(C) Both Caller and Callee

(D) None of the other options

Ans : b

6.

"In a callee-save register convention, who is responsible for saving registers?"

(A) Caller
(B) Callee
(C) Both Caller and Callee
(D) None of the above
"In a callee-save register convention, who is responsible for saving registers?"

Ans : b

7.

Local storage is created by

(A) Callee

(B) Caller

(C) Both Caller and Callee

(D) None of the other options

Ans : a

8.

For a switch statement, the expression can result into values in the range -5 to +6. Number of entries in the jumptable should be

(A) 5

(B) 6

(C) 11

(D) 12

Ans : d

9.

For a switch statement implemented as a jumptable, default_case is

(A) A part of jumptable

(B) Not a part of jumptable

(C) in the middle of the jumptable

(D) at the beginning of the jumptable

Ans : b

10.

For pair of goto based storage allocation for functions, the second goto statement transfers control to the beginning of

(A) Storage space

(B) Function code

(C) Program

(D) None of the other options

Ans : b

11.

In a jump table-based switch statement, how is the index for the jump table calculated when the case values include negative numbers?

(A) Using the case value directly as the index
(B) By shifting all case values so that the smallest value maps to index 0
(C) By skipping negative case values in the jump table
(D) By using absolute values of case numbers

Ans : b

12.

Which of the following occurs during the calling sequence of a function call?

(A) Saving the return address and passing arguments
(B) Restoring registers and returning control to the caller
(C) Freeing allocated memory and clearing local variables
(D) None of the above

Ans : a